

### 3.5.4 Nutrient cycles (A-level only)

| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Opportunities for skills development                                                                                |
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| <p>Nutrients are recycled within natural ecosystems, exemplified by the nitrogen cycle and the phosphorus cycle.</p> <p>Microorganisms play a vital role in recycling chemical elements such as phosphorus and nitrogen.</p> <ul style="list-style-type: none"><li>• The role of saprobionts in decomposition.</li><li>• The role of mycorrhizae in facilitating the uptake of water and inorganic ions by plants.</li><li>• The role of bacteria in the nitrogen cycle in sufficient detail to illustrate the processes of saprobiotic nutrition, ammonification, nitrification, nitrogen fixation and denitrification.</li></ul> <p>(The names of individual species of bacteria are <b>not</b> required).</p> <p>The use of natural and artificial fertilisers to replace the nitrates and phosphates lost by harvesting plants and removing livestock.</p> <p>The environmental issues arising from the use of fertilisers including leaching and eutrophication.</p> | <p><b>PS 1.1</b></p> <p>Students could devise investigations into the effect of named minerals on plant growth.</p> |